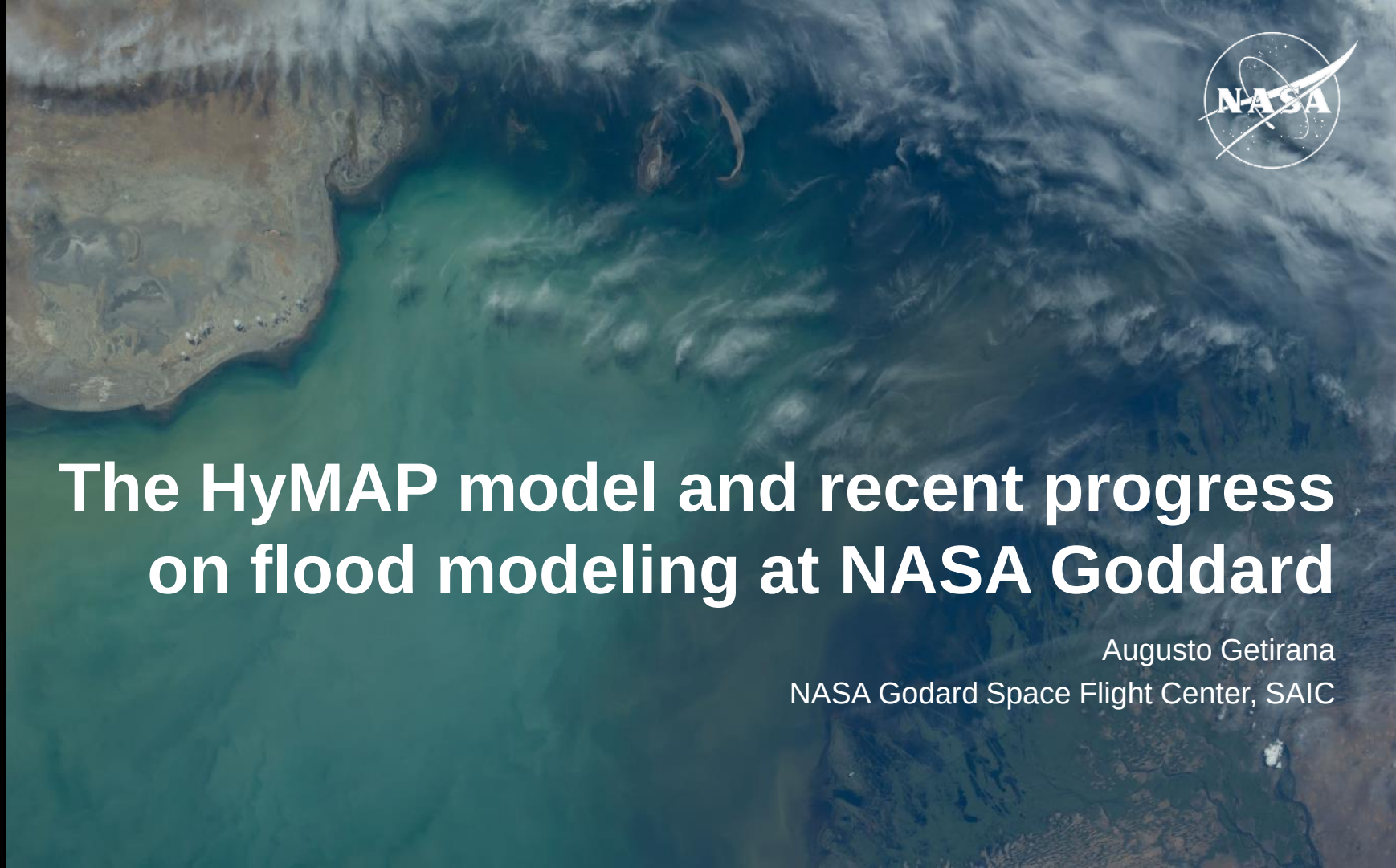


The HyMAP model and recent progress on flood modeling at NASA Goddard

Augusto Getirana
NASA Godard Space Flight Center, SAIC

Why should we care about flood modeling?

Climate change and human activities have been magnifying the occurrence, intensity and impacts of extreme hydrological events



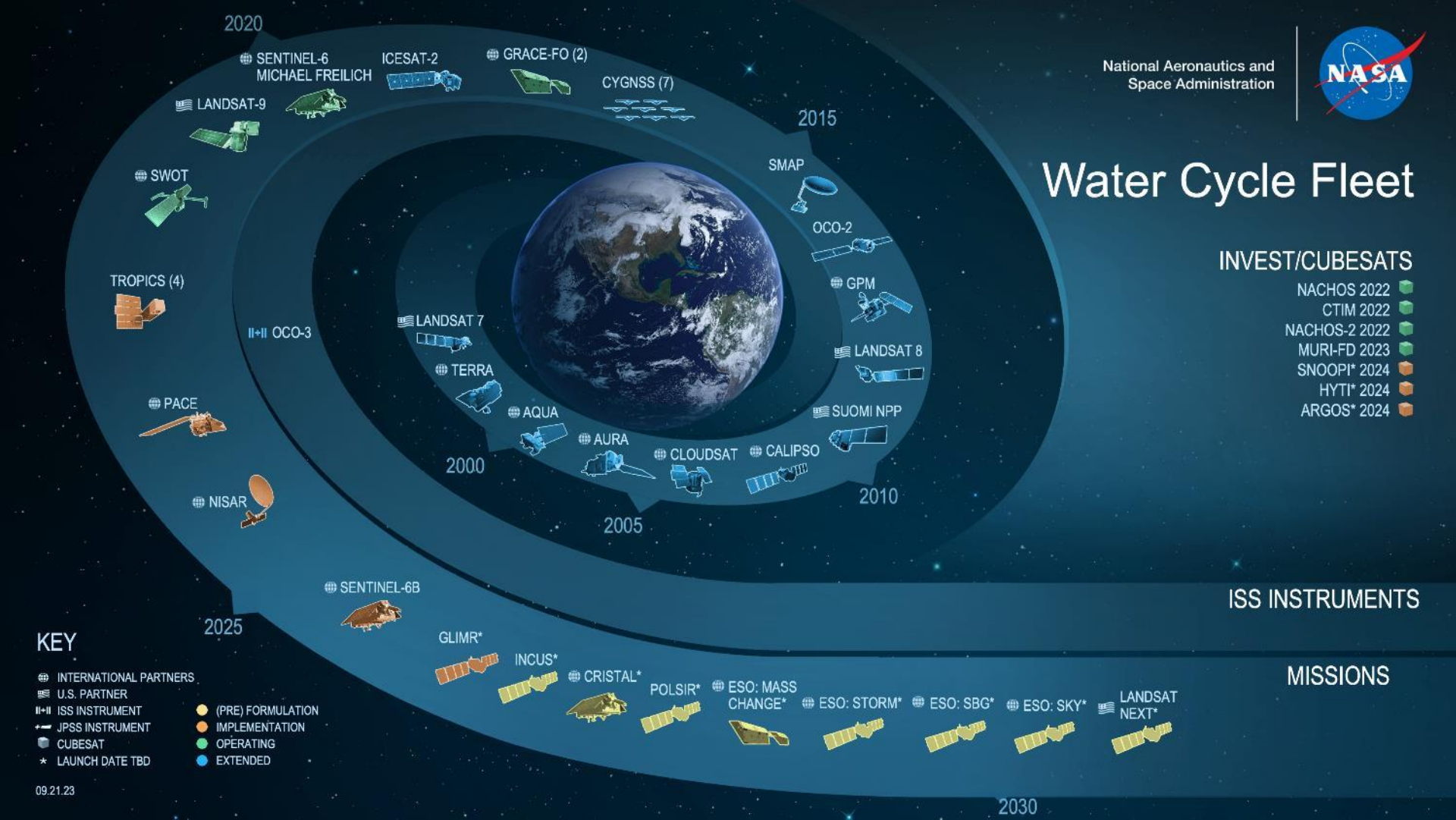











National Aeronautics and Space Administration
NASA
Water Cycle Fleet
2020
SENTINEL-6 MICHAEL FREILICH
ICESAT-2
GRACE-FO (2)
CYGNSS (7)
LANDSAT-9
SWOT
TROPICS (4)
PACE
NISAR
SENTINEL-6B
2025
2015
SMAP
OCO-2
GPM
LANDSAT 8
SUOMI NPP
CALIPSO
2010
2005
2000
OCO-3
LANDSAT 7
TERRA
AQUA
AURA
CLOUDSAT
INVEST/CUBESATS
NACHOS 2022
CTIM 2022
NACHOS-2 2022
MURI-FD 2023
SNOOPI* 2024
HYTI* 2024
ARGOS* 2024
ISS INSTRUMENTS
MISSIONS
GLIMR*
INCUS*
CRISTAL*
POLSIR*
ESO: MASS CHANGE*
ESO: STORM*
ESO: SBG*
ESO: SKY*
LANDSAT NEXT*
2030
KEY
INTERNATIONAL PARTNERS
U.S. PARTNER
ISS INSTRUMENT
JPSS INSTRUMENT
CUBESAT
LAUNCH DATE TBD
(PRE) FORMULATION
IMPLEMENTATION
OPERATING
EXTENDED
09.21.23

Remote sensing of surface hydrology

Advantages

- Monitoring of remote regions;
- Spatial coverage;
- Open access.

Disadvantages

- Temporal resolution;
- Limited physical processes;
- Relatively short time series;
- Past events only.

Integration with computational models

- Evaluation;
- Calibration;
- Data assimilation.



Result

- Estimates of unobserved physical processes;
- Refined spatial and temporal resolutions;
- Scenarios;
- Forecasts.

GODDARD EARTH SCIENCES

Flood models and river routing schemes

Main characteristics

- Spatial and temporal scales;
- Discretization;
- Complexity of physical processes.

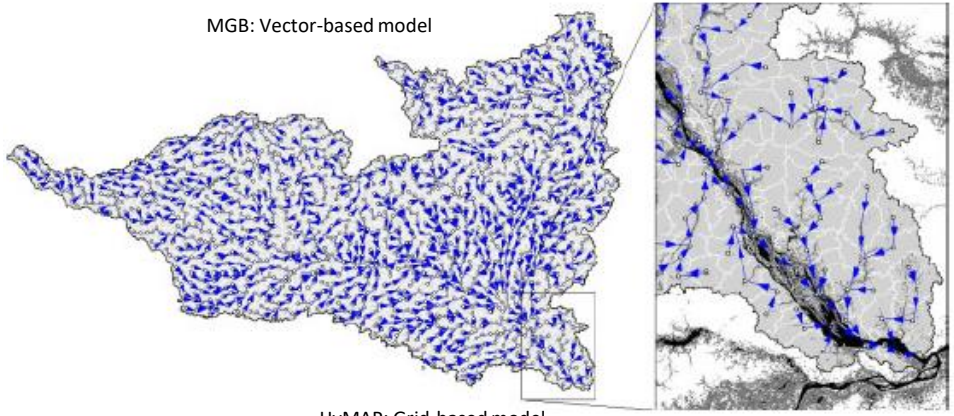
Global scale

TRIP (Oki et al., 1999)
WaterGAP (Doll et al., 2003)
CTRIP (Decharme et al., 2011)
CaMa-Flood (Yamazaki et al., 2011)
HyMAP (Getirana et al., 2012)
MOSART (Li et al., 2013)
DRTR (Wu et al., 2013)
RAPID (David et al., 2011)

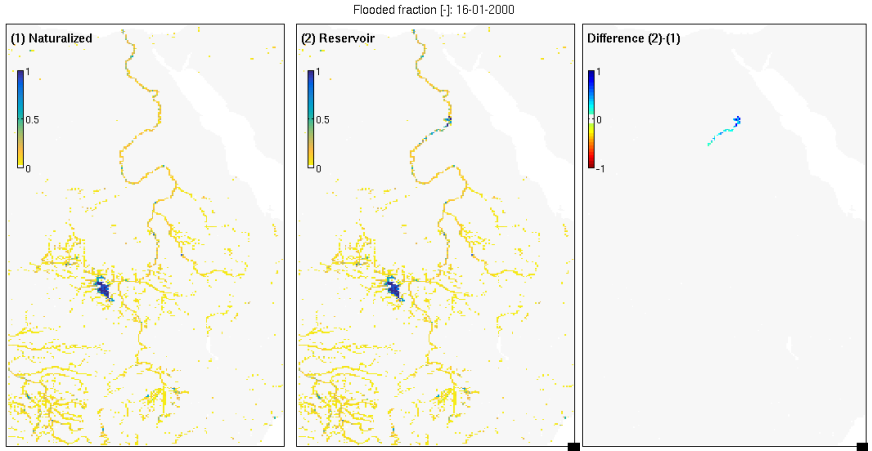
Continental/meso scale

LISFLOOD-FP (Bates and De Roo, 2000)
MGB (Collischonn et al., 2007)
LEAF-Hydro-Flood (Miguez-Macho et al., 2007)
THMB (Coe et al., 2008)

MGB: Vector-based model



HyMAP: Grid-based model



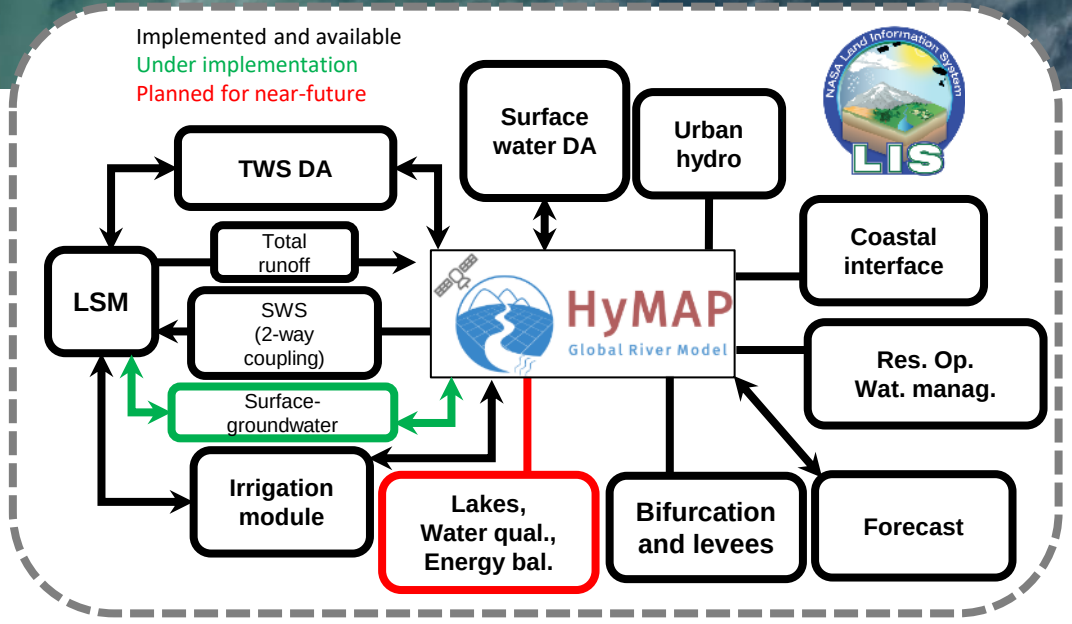
LIS surface hydrology capabilities

HyMAP Features

Implemented:
- Local inertia formulation
- Adaptive time step
- Backwater effects
- Floodplain dynamics
- 2-way coupling
- Reservoir operation
- Water management
- GRACE-DA correction for SWS
- Surface water DA (water level, discharge and extent)
- Urban flood and drainage
- Coastal interface
- Bifurcation
- Irrigation coupling

Ongoing:
- Surface-groundwater coupling

Future:
- Sediment dynamics
- Lake module

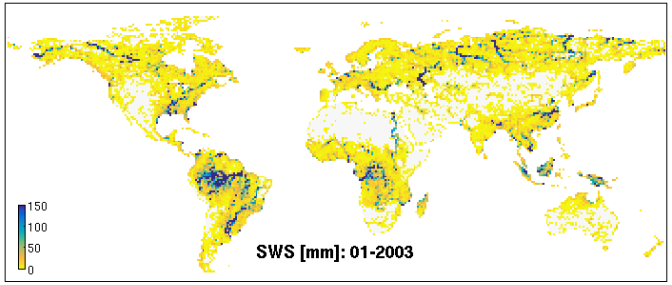


Outputs

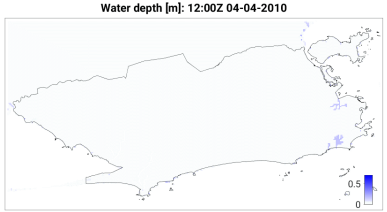
- Water discharge
- Water depth
- Water storage
- Flow velocity
- Floodplain extent
- River slope

Flexible spatial scale

1° global surface water storage variability



200-m streamflow over Rio de Janeiro city





HyMAP

Global River Model

K water 한국수자원공사

U.S. AIR FORCE

INPE

CILSS

JOHNS HOPKINS UNIVERSITY

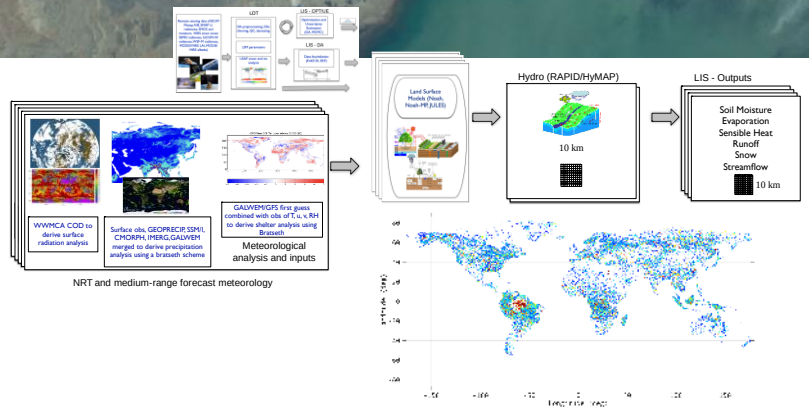
FEWS NET
FAMINE EARLY WARNING SYSTEMS NETWORK

NASA

RIO PREFEITURA

GODDARD EARTH SCIENCES

Examples of application environments that employ HyMAP



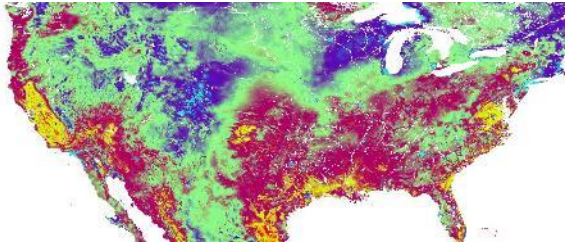
DoD/IC environments

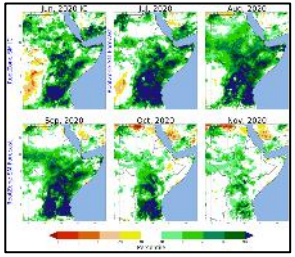
U.S. Air Force Weather employs an operational global LIS configuration (at 10km spatial resolution), which was extended to include HyMAP in 2021

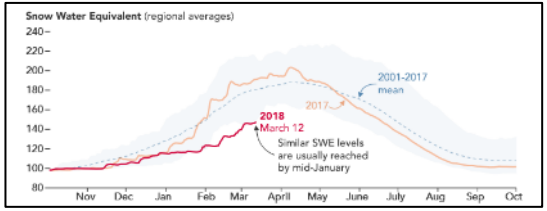
This configuration already includes the use of SMAP and GPM data. The operational LIS-HyMAP system will enable the future transition of SWOT data in operations.

High-resolution North American LDAS

A high resolution (~1km) LDAS is being developed over CONUS/North America that: 1) enables the assimilation of NASA remote sensing assets (GPM, SMAP, MODIS, etc.), 2) allows the incorporation of future high resolution remote sensing measurements (NISAR, SWOT). This environment will function as a research testbed for evaluating the utility of remote sensing measurements







Food security LDAS

NASA Famine Early Warning Systems Network (FEWSNET) environments employ LIS-based modeling and assimilation systems. This configuration can also be used to demonstrate the use of future remote sensing observations.

Content of this seminar

- Rivers and floodplains as key components of terrestrial water storage variability
- Climate and human impacts on flood risk in Southern Louisiana
- Development of a flood model for Rio de Janeiro city, Brazil
- **Final considerations and prospects**

Content of this seminar

- Rivers and floodplains as key components of terrestrial water storage variability
- Climate and human impacts on flood risk in Southern Louisiana
- Development of a flood model for Rio de Janeiro city, Brazil
- Final considerations and prospects

Rivers and floodplains as key components of terrestrial water storage variability



Geophysical Research Letters

RESEARCH LETTER
10.1002/2017GL074684

Rivers and Floodplains as Key Components of Global Terrestrial Water Storage Variability

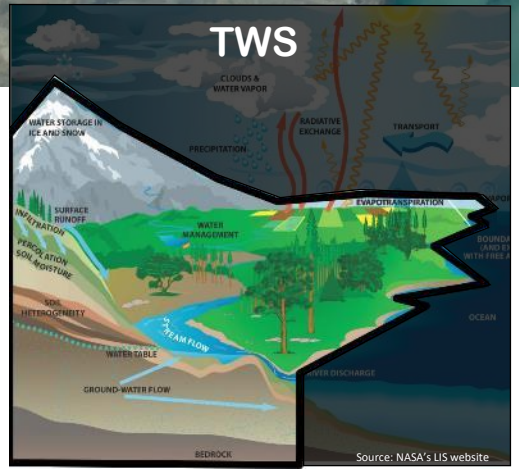
Augusto Getirana[1,2], Sujay Kumar[1], Manuela Girotto[3,4], and Matthew Rodell[1]

Science questions

1. What is the global impact of surface water storage on TWS?
2. How does that impact vary spatially?

GODDARD EARTH SCIENCES

Modeling Framework



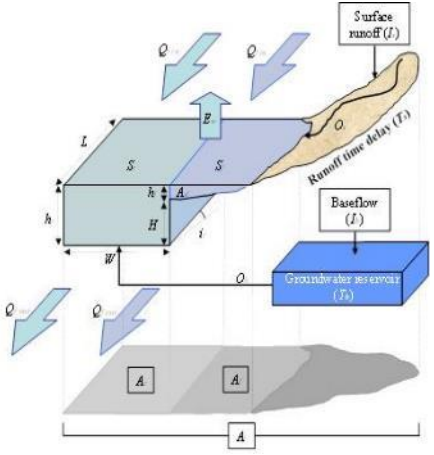
NASA Land Information System

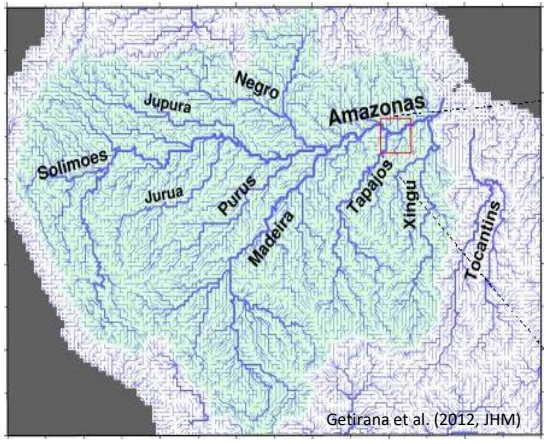
LIS

Surface water dynamics;
Surface water storage (SWS)

Vertical water and energy balance;
Land water storage (LWS)

HyMAP



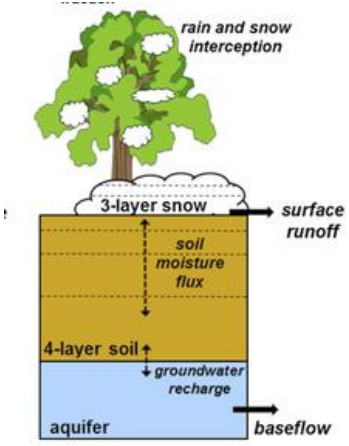


Getirana et al. (2012, JHM)

Surface runoff

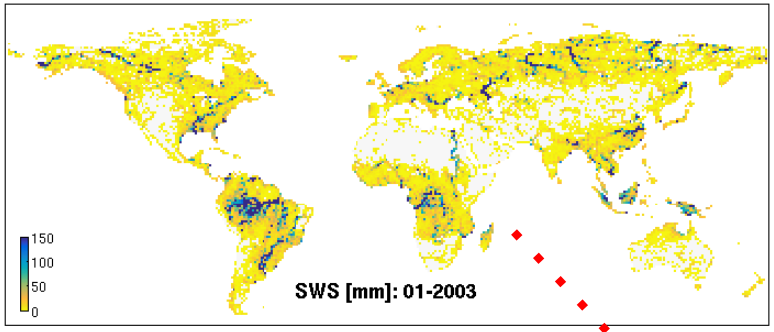
Baseflow

Noah-MP

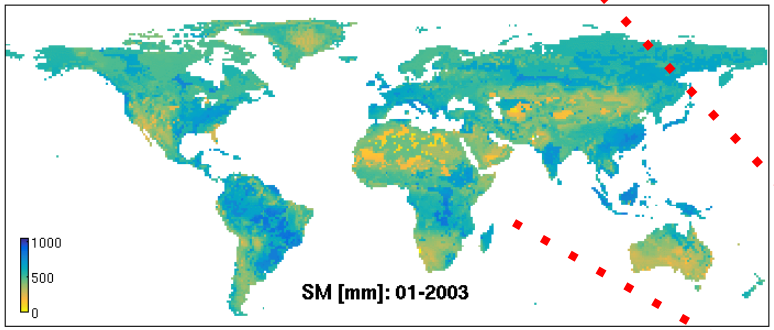


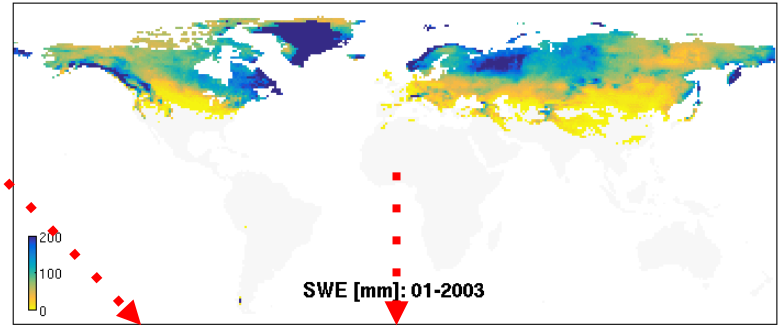
Mendoza et al. (2015, JHM)

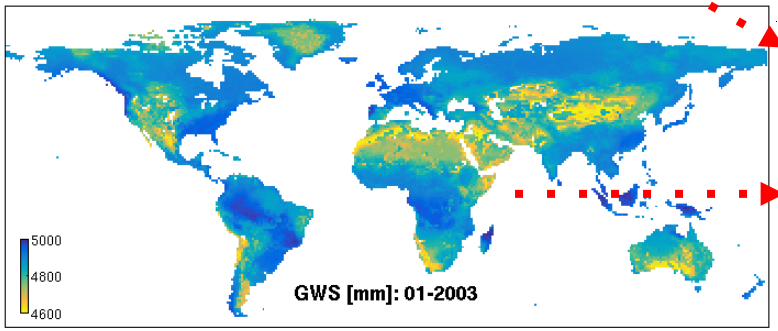
GLOBAL TERRESTRIAL WATER STORAGE

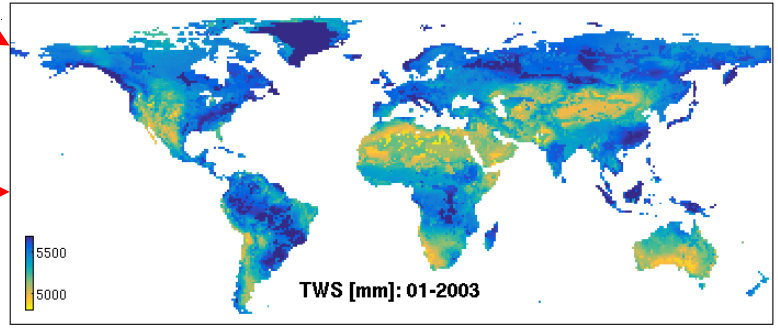


- About 2,860km^3 of water is stored in rivers and floodplains [previous estimates vary from 2,000km^3 (Oki & Kanae, 2006) to 2,120km^3 (Shiklomanov, 1993)];
- 28% of that water, or 800km^3, is concentrated in the Amazon Basin;
- Uncertainty due to meteorological forcings is about 6% globally.

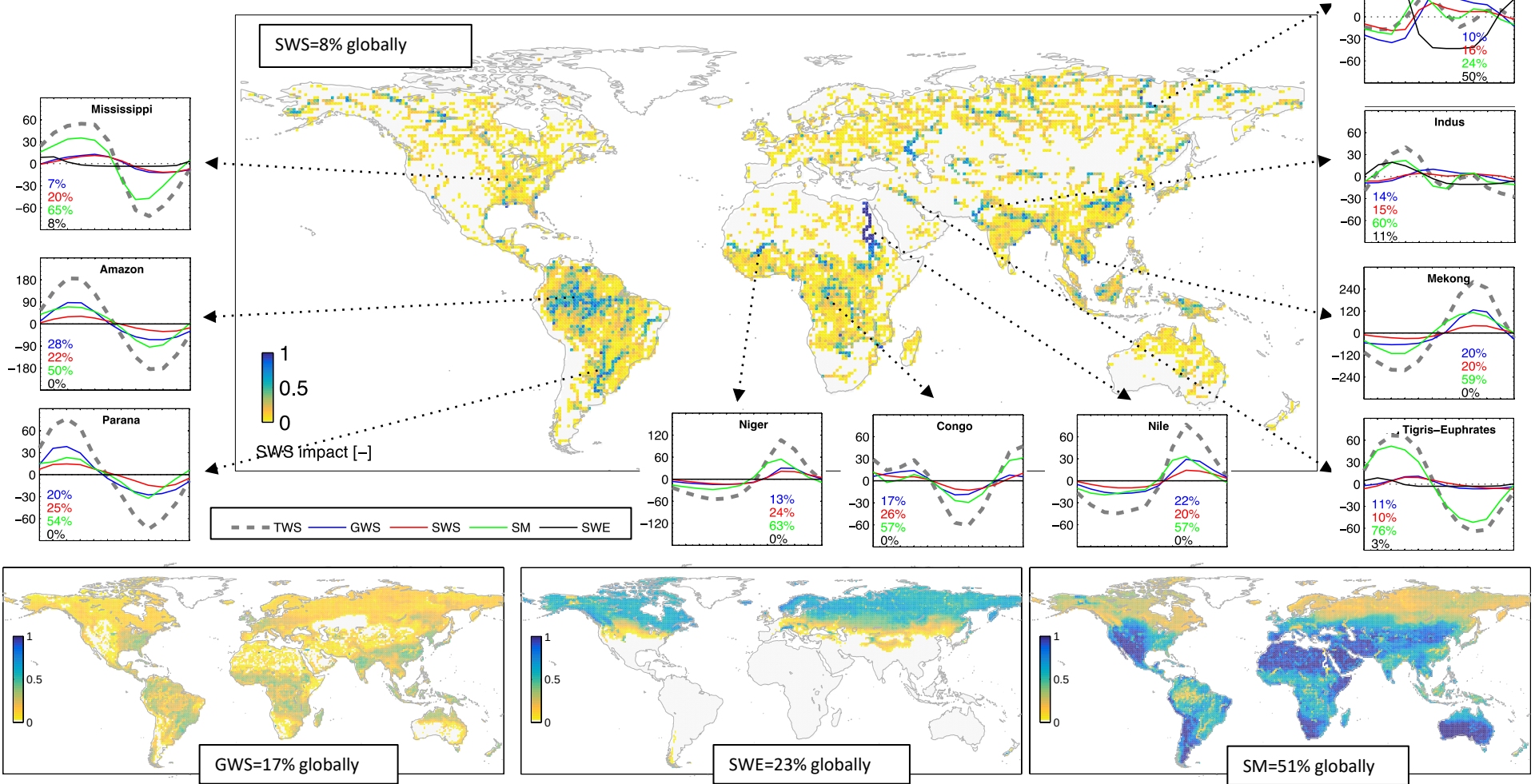








IMPACTS ON TWS

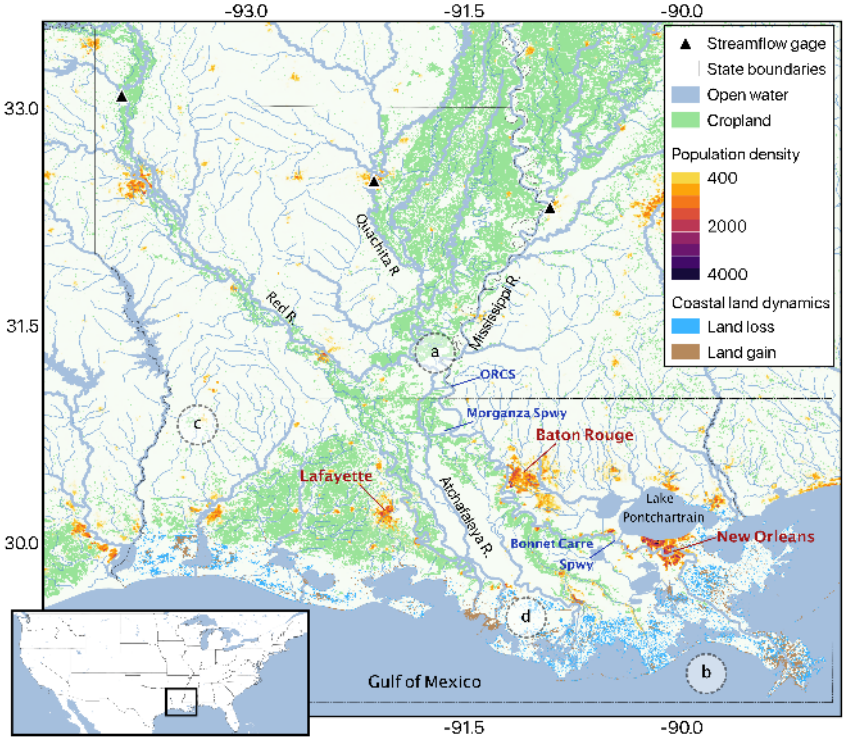


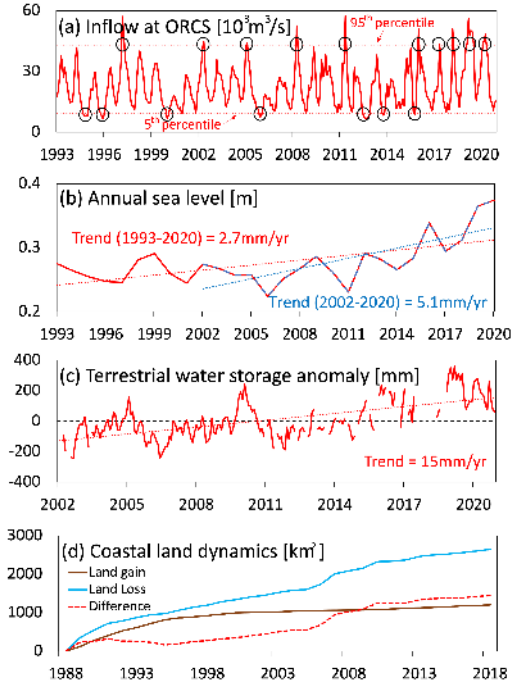
Content of this seminar

- Climate and human impacts on flood risk in Southern Louisiana
- Development of a flood model for Rio de Janeiro city, Brazil
- Final considerations and prospects

Climate and human impacts on flood risk in Southern Louisiana

Observation-based evidence of hydrological change, sea level rise, increased terrestrial water storage and land loss





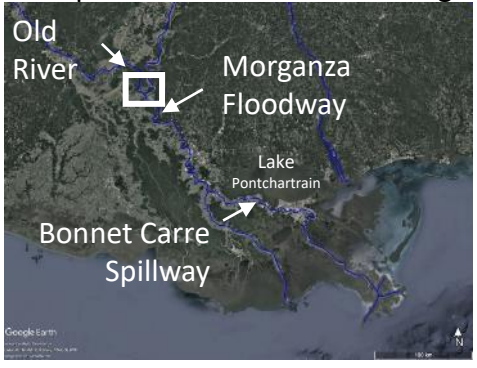
GODDARD EARTH SCIENCES

Model configuration

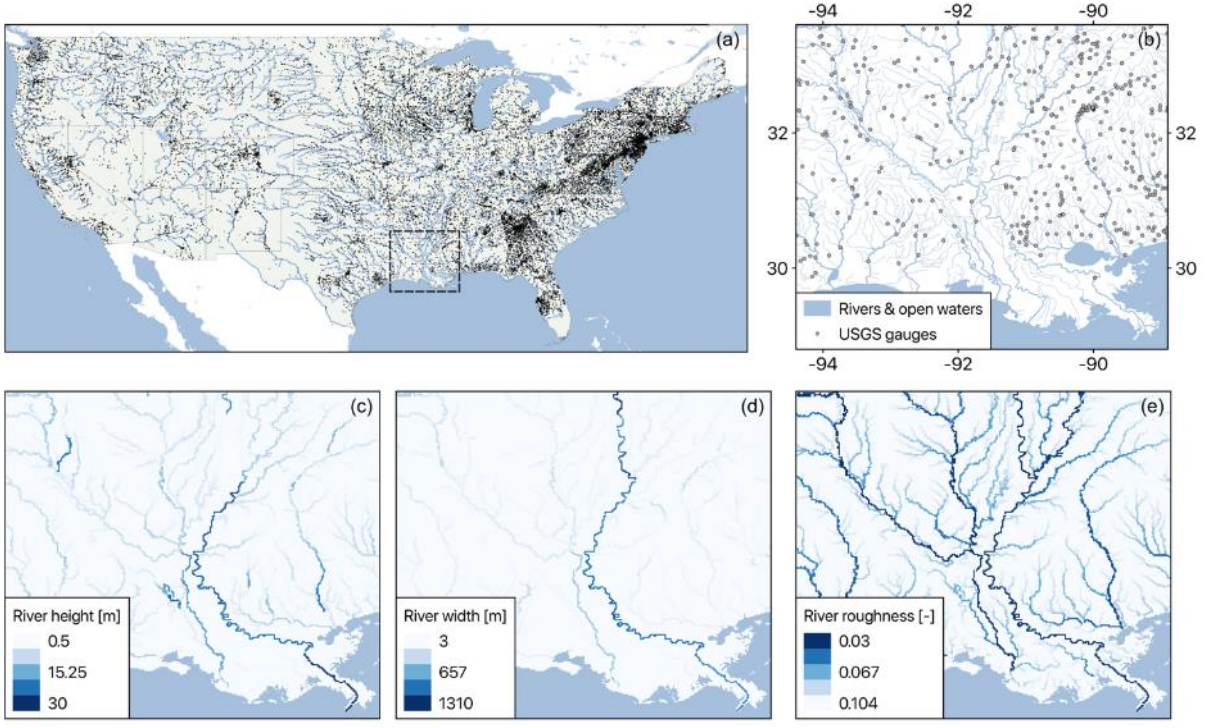
- 1993-2020 period;
- 2km spatial resolution;
- MERRA-2;
- Streamflow obs used as upstream boundary condition at 3 locations;
- Radar altimetry used as downstream boundary condition;
- 4 experiments;

Scenario	Hydrology	Sea level	Water management
1	Yes	7-day	Yes
2	Yes	7-day	No
3	Yes	Climatology	No
4	Yes	Constant	No

- Representation of water management

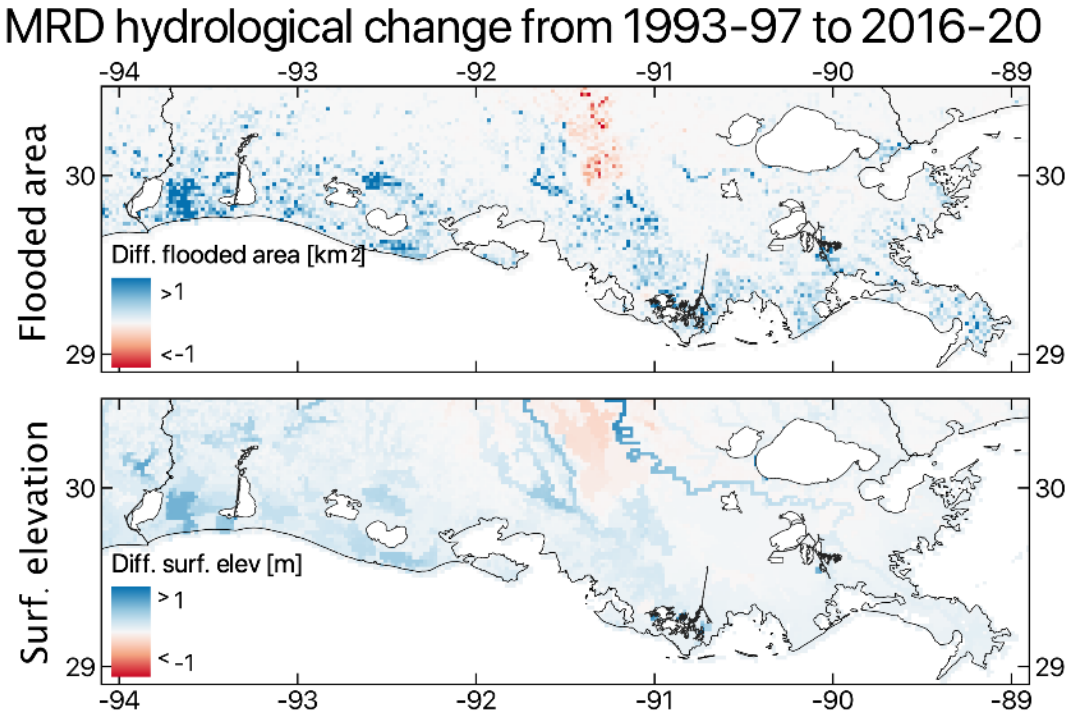


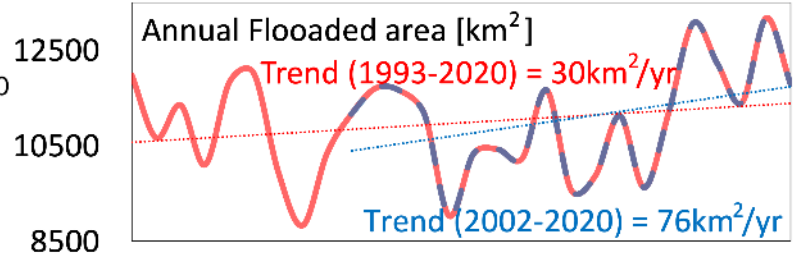
River geometry parameterization

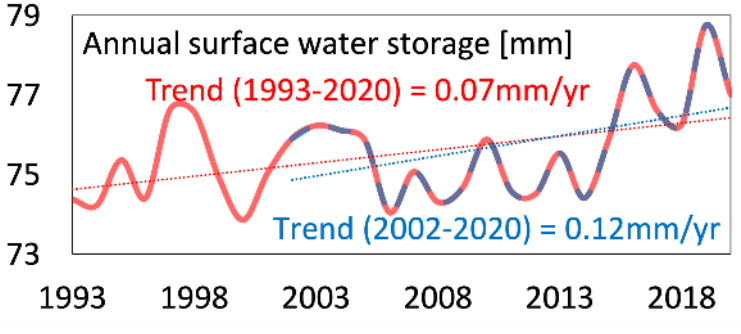


Model-based MRD hydrological changes

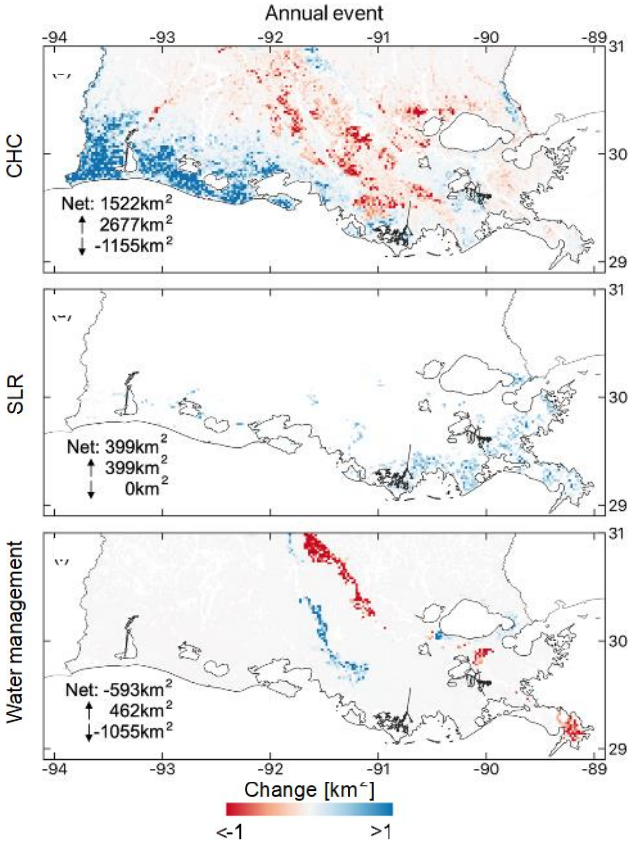
Based on Scenario 1 (i.e., all factors considered), surface elevation and flooded area generally increase from 1993-97 to 2016-20





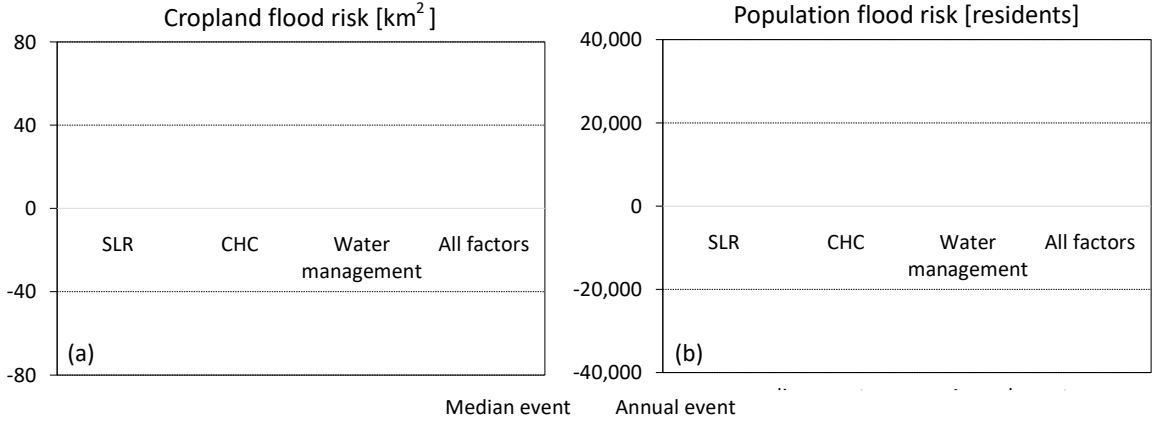


Coastal interface & water management



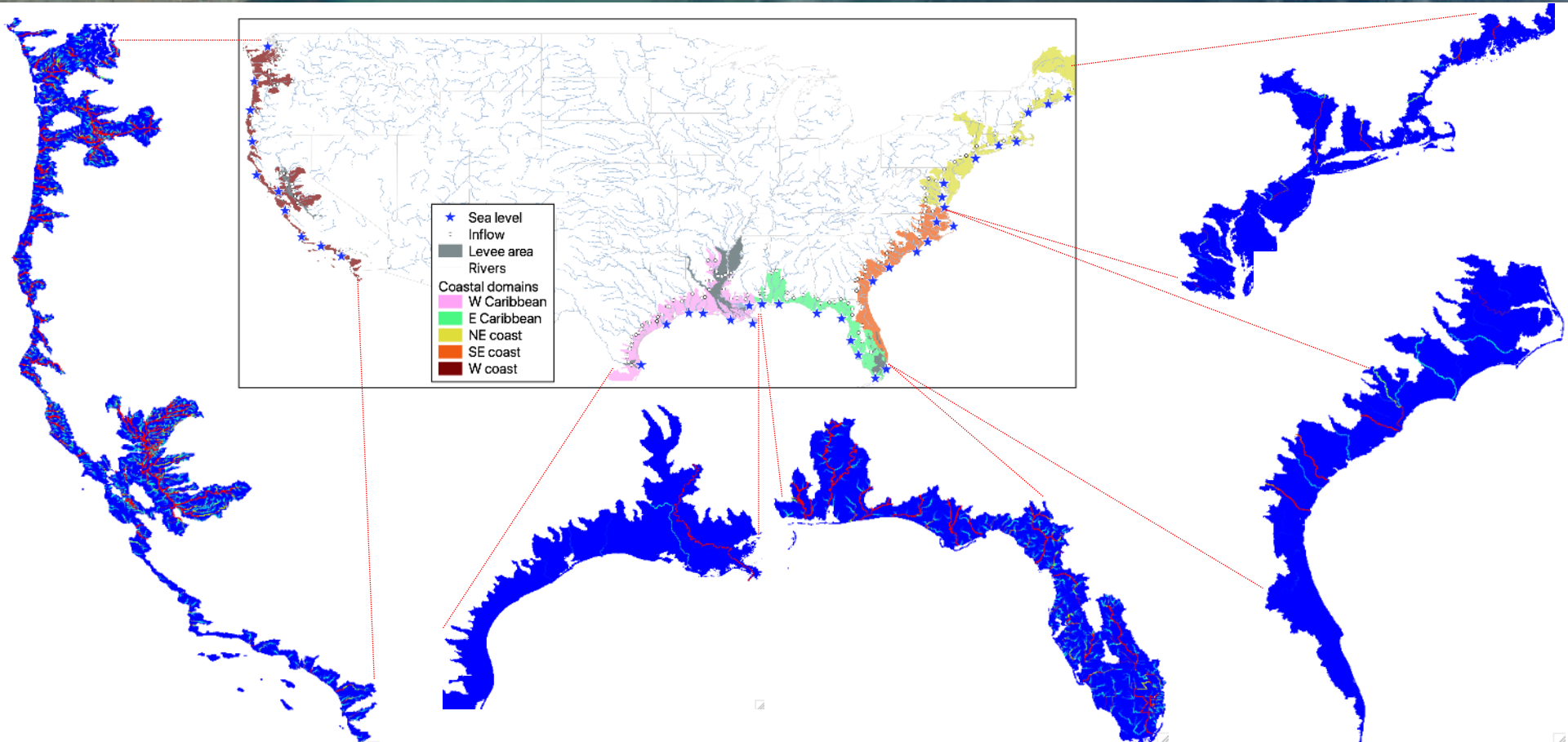
Changes in flooded areas due to CHC, SLR, and water management over coastal Louisiana for annual events

3 factors considered in our modeling system: Climate-induced hydrological changes (CHC), sea level rise (SLR) and water management



Flood risk over coastal Louisiana as a function of return period (median and annual events) due to SLR, CHC, water management and all factors combined (all factors).

Next step: all U.S. coasts

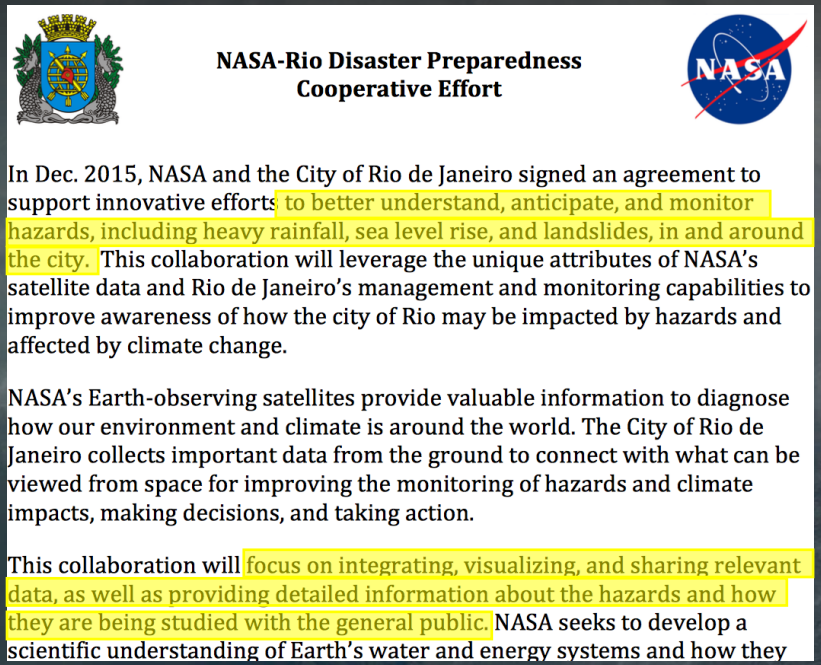


Content of this seminar

- Rivers and floodplains as key components of terrestrial water storage variability
- Climate and human impacts on flood risk in Southern Louisiana
- Development of a flood model for Rio de Janeiro city, Brazil
- Final considerations and prospects

NASA-Rio de Janeiro Partnership

- Better understand, anticipate and monitor **environmental hazards** in Rio de Janeiro
- **Pioneer applications** of NASA Earth Observations at the urban scale
- Conduct **joint activities** that leverage NASA's remote sensing products and Rio de Janeiro's monitoring and crisis management capabilities
- Bring NASA Earth science to **Brazilian educators** and students



NASA-Rio Disaster Preparedness Cooperative Effort

In Dec. 2015, NASA and the City of Rio de Janeiro signed an agreement to support innovative efforts to better understand, anticipate, and monitor hazards, including heavy rainfall, sea level rise, and landslides, in and around the city. This collaboration will leverage the unique attributes of NASA's satellite data and Rio de Janeiro's management and monitoring capabilities to improve awareness of how the city of Rio may be impacted by hazards and affected by climate change.

NASA's Earth-observing satellites provide valuable information to diagnose how our environment and climate is around the world. The City of Rio de Janeiro collects important data from the ground to connect with what can be viewed from space for improving the monitoring of hazards and climate impacts, making decisions, and taking action.

This collaboration will focus on integrating, visualizing, and sharing relevant data, as well as providing detailed information about the hazards and how they are being studied with the general public. NASA seeks to develop a scientific understanding of Earth's water and energy systems and how they

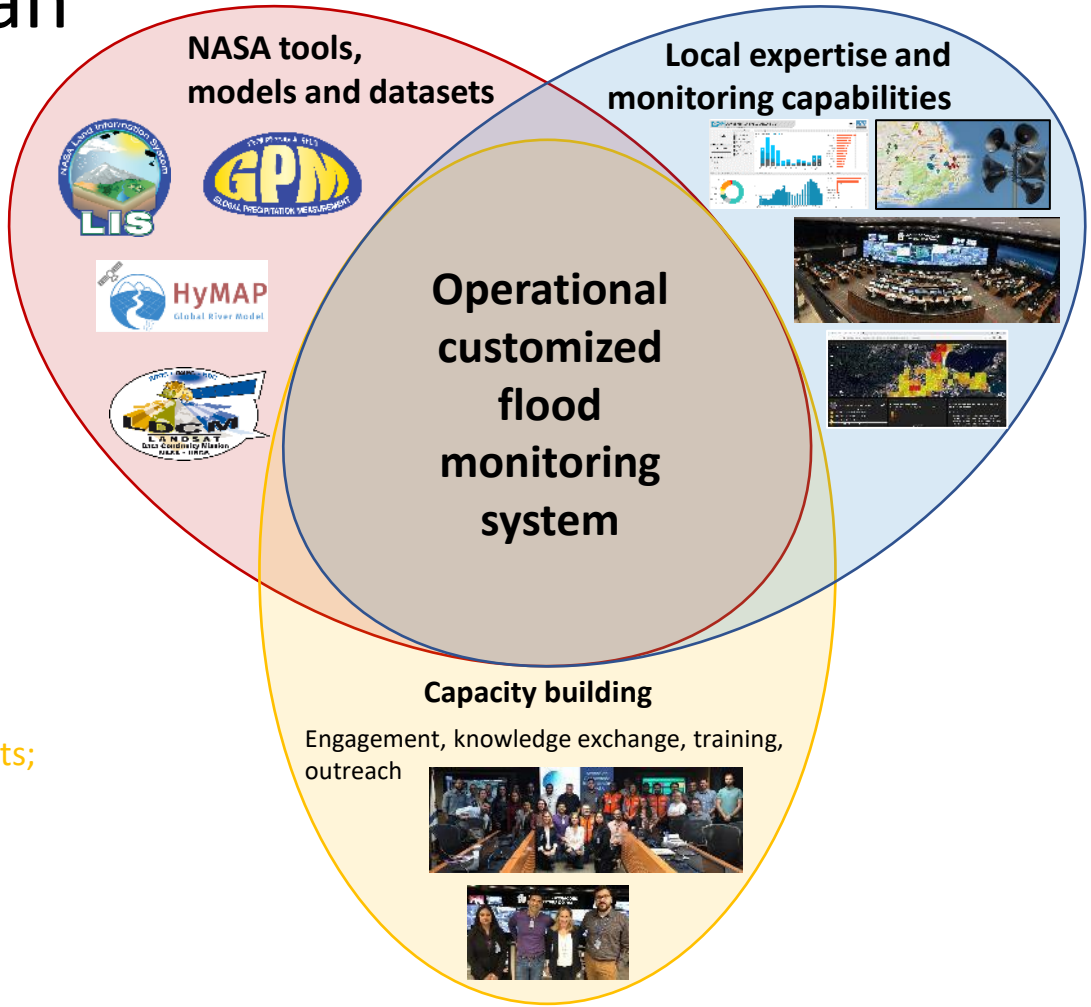
Courtesy: D. Kirschbaum

Goal and proposed plan

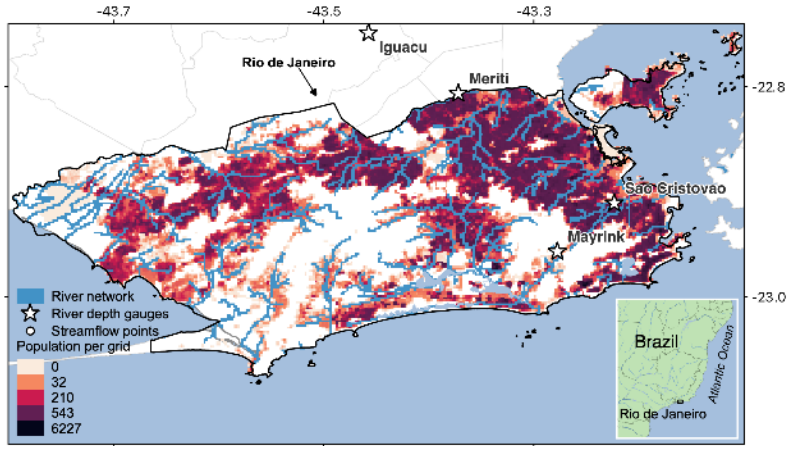
- Land Information System (LIS);
- Land surface Data Toolkit (LDT);
- GPM;
- Landsat;
- Modeling and satellite expertise.

- Real-time hydrometeorological monitoring;
- Local land use and soil type distribution;
- Lidar-based DEM;
- River network and geometry;
- Flood mapping;
- Urban drainage system, etc.

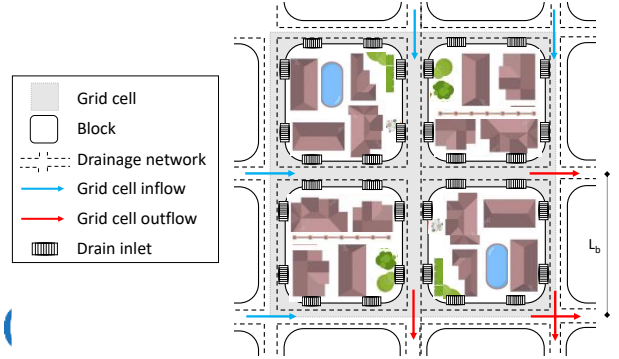
- Frequent phone calls, emails;
- Updates on results, challenges and requirements;
- Annual in person meetings and workshops;
- Strong local interest and technical expertise;
- Easy and immediate access to information, etc.

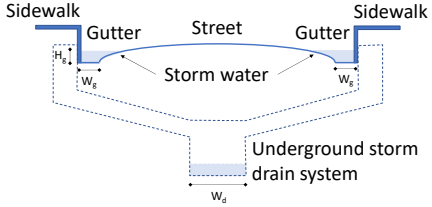


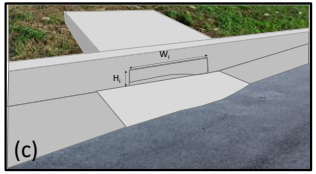
HyMAP's urban drainage module



(a) (b)





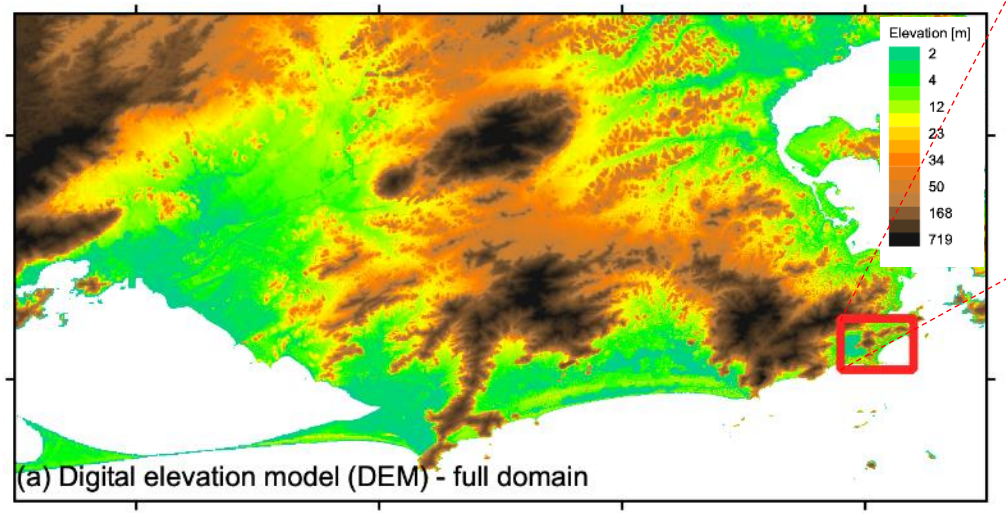


Drainage network parameters	
Street block length [m]	100
Underground squared pipe width [m]	1
Drainage length density [m/m^2]	0.02
Roughness coefficient for cement pipes [-]	0.012
Drainage system slope [m/m]	0.003

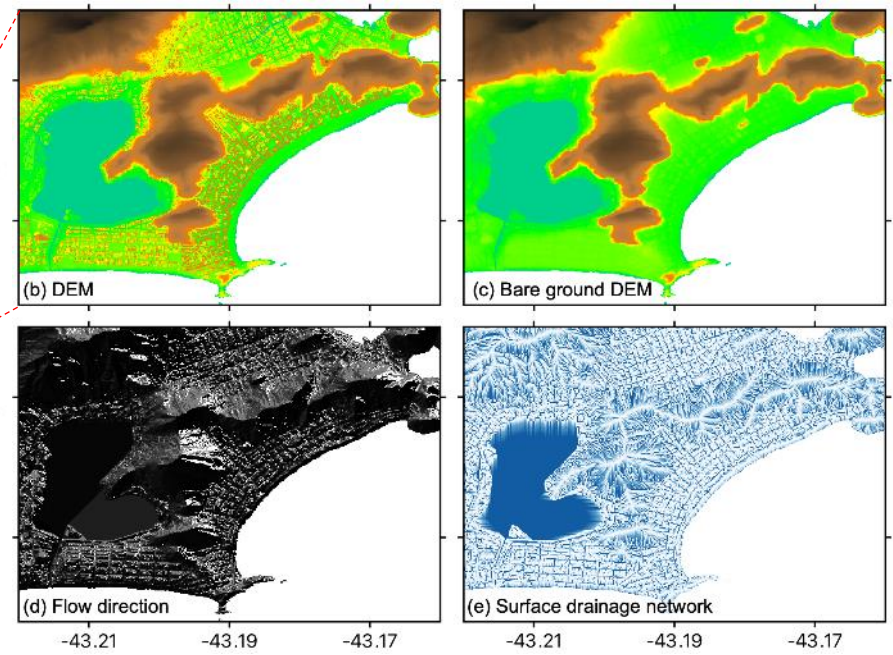
Gutter parameters	
Gutter width [m]	1
Gutter height [m]	0.15
Gutter density [units/m^2]	0.0008
Gutter water intake velocity [m/s]	2

Topography data processing

Lidar data (8 pts/m^2) => 10x10m^2
+
MERIT DEM (90x90m)



Hydrological information acquisition

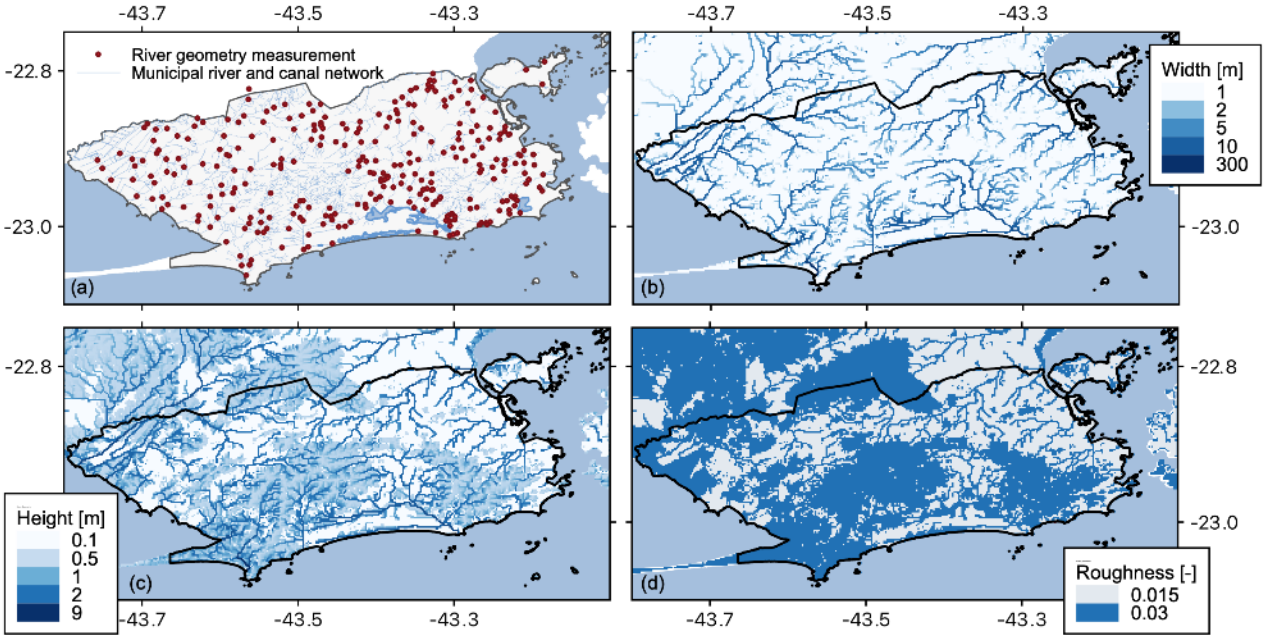


River network and geometry

River geometry collected by Rio Aguas
+
Manual measurements on Google Earth

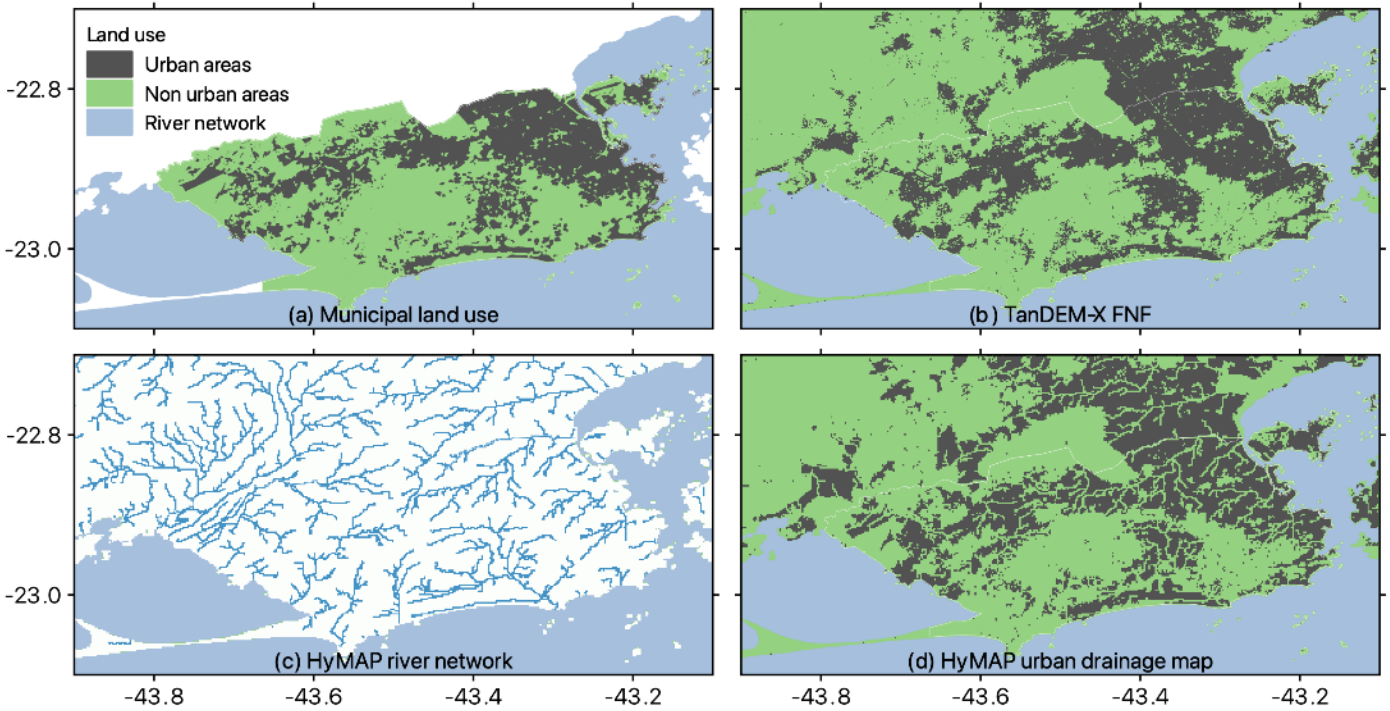
Total of >300 points with river width and/or depth

Spatial distribution using (i) linear interpolation and (ii) functions with drainage area



Urban drainage mapping

Municipal land use classification
+
Global land use classification
+
River network

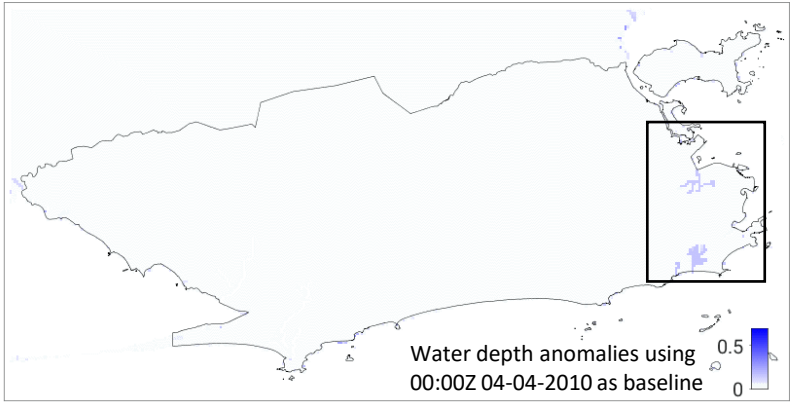


One of the most severe floods ever recorded

Rainfall [mm/h]: 12:00Z 04-04-2010

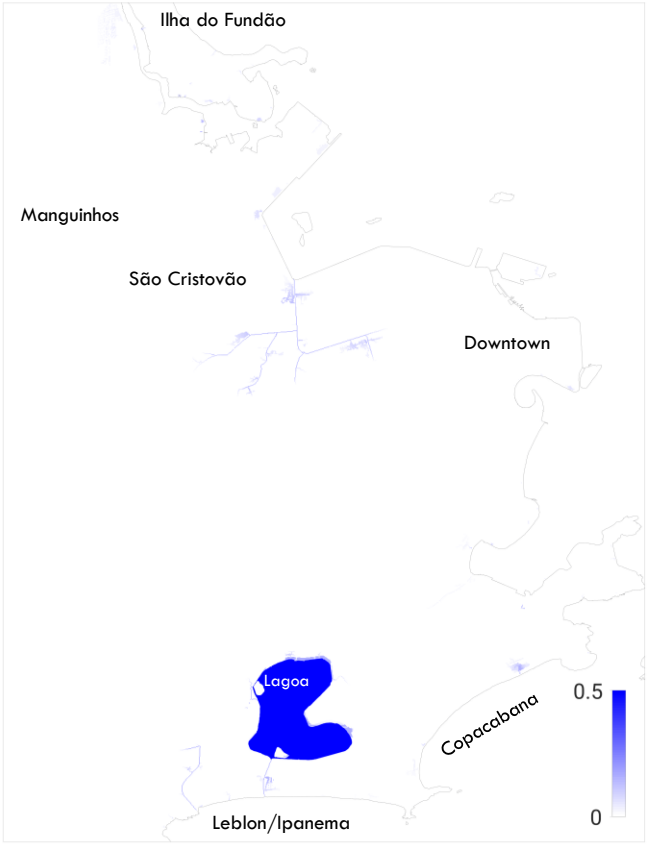


Water depth [m]: 12:00Z 04-04-2010



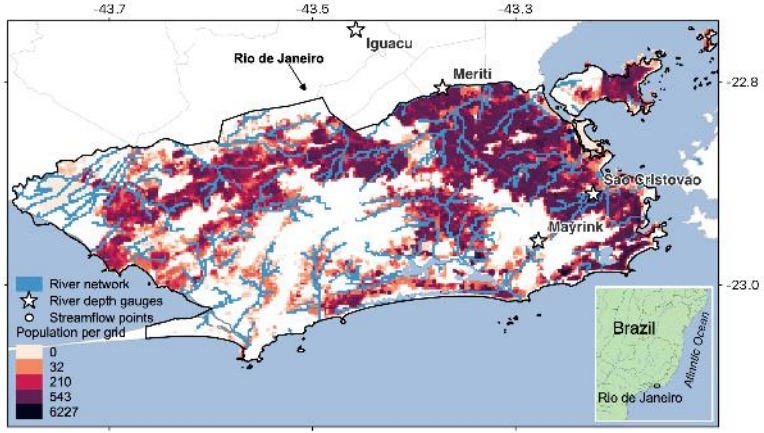
Downscaled water depth anomalies at 10-meter spatial resolution during major flood occurred in April 2010

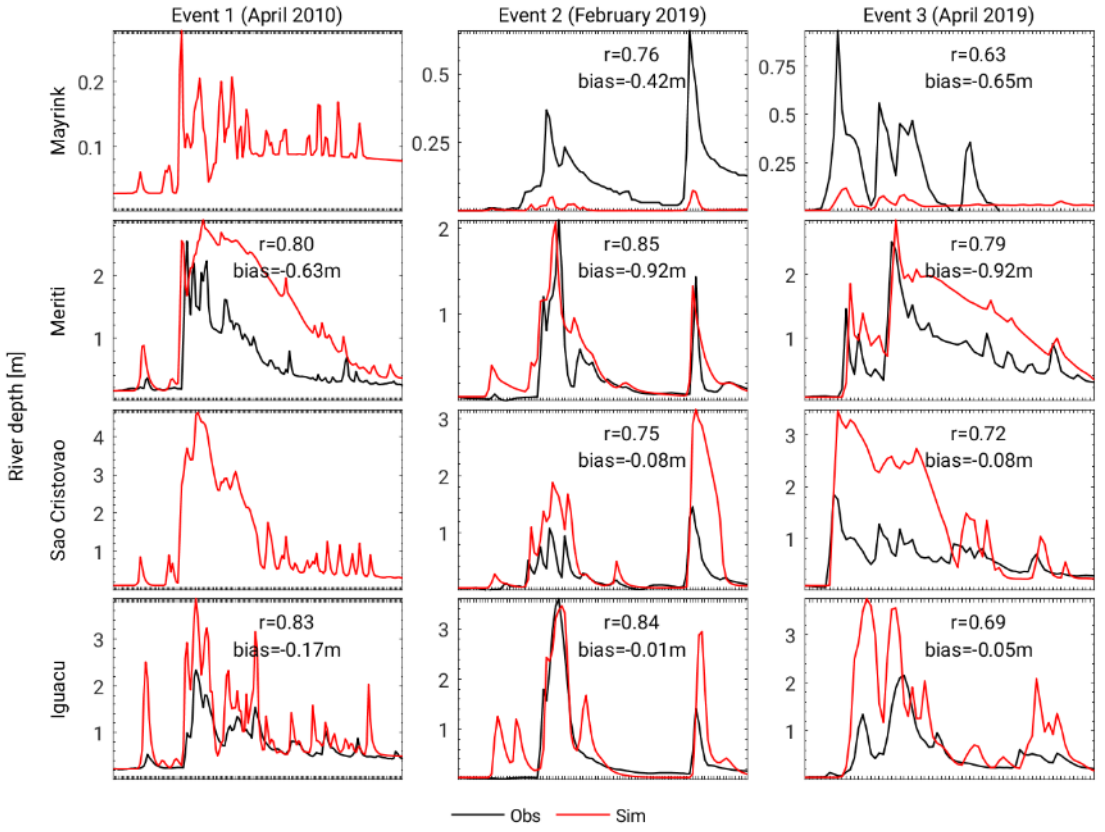
Water depth [m]: 12:00Z 04-04-2010



Model evaluation

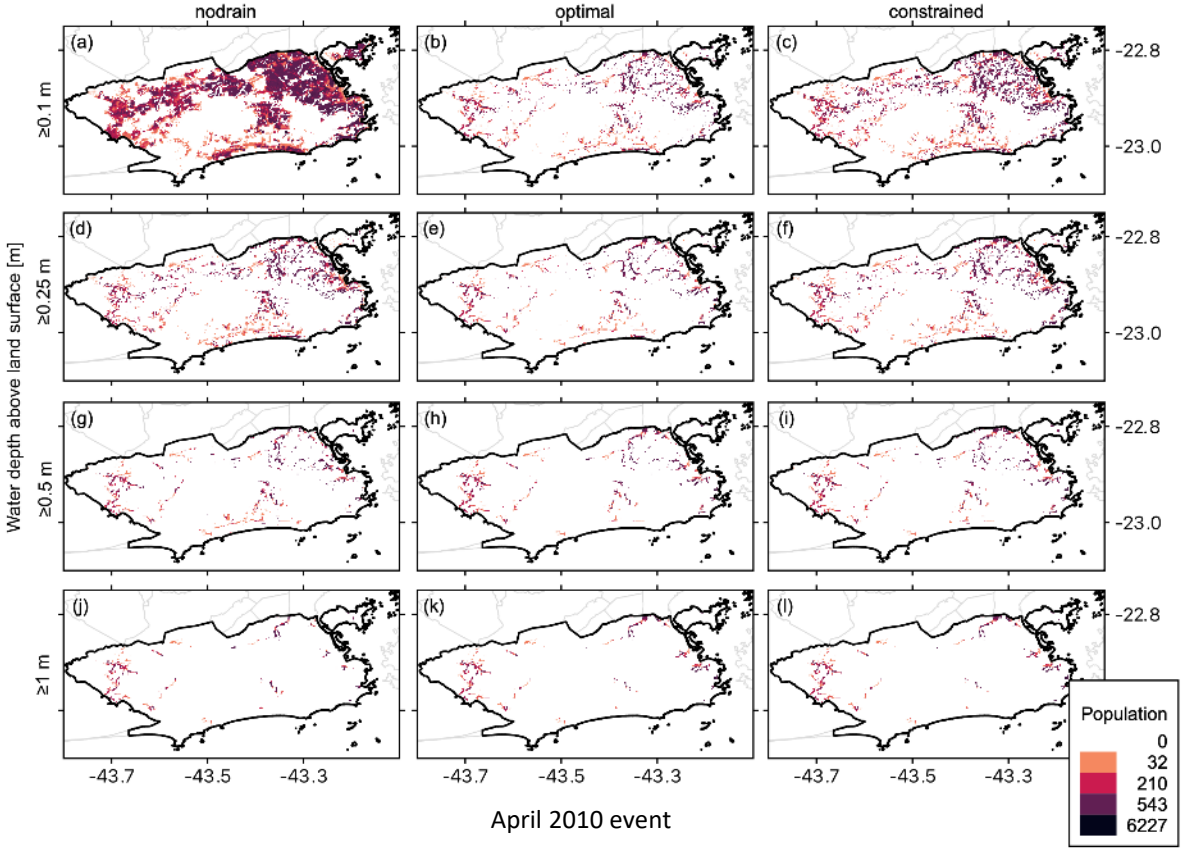
- Water level evaluation at 4 INEA gauge stations
- 3 major flood events in the city
- Overall good results
- The model detected flood timing and magnitude





Mapping population exposed to flooding

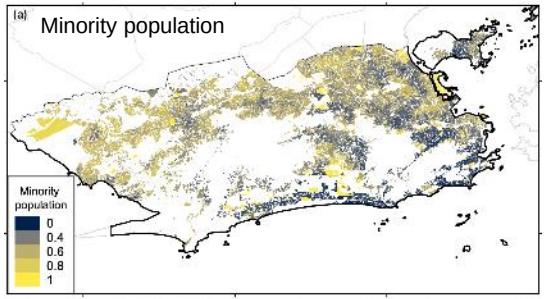
- Using census data, we can identify populations exposed to extreme flooding
- 3 experiments with different drainage systems
- This show how this toll can be used in urban planning

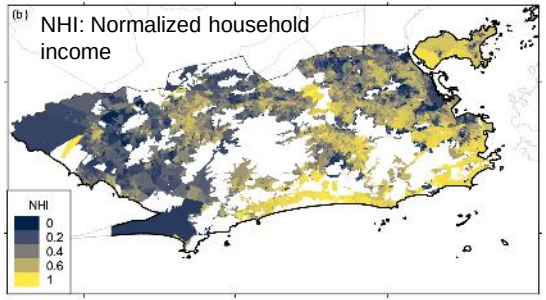


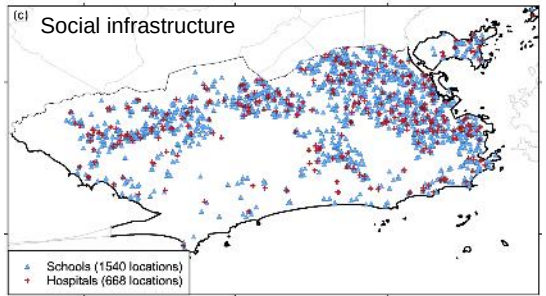
April 2010 event

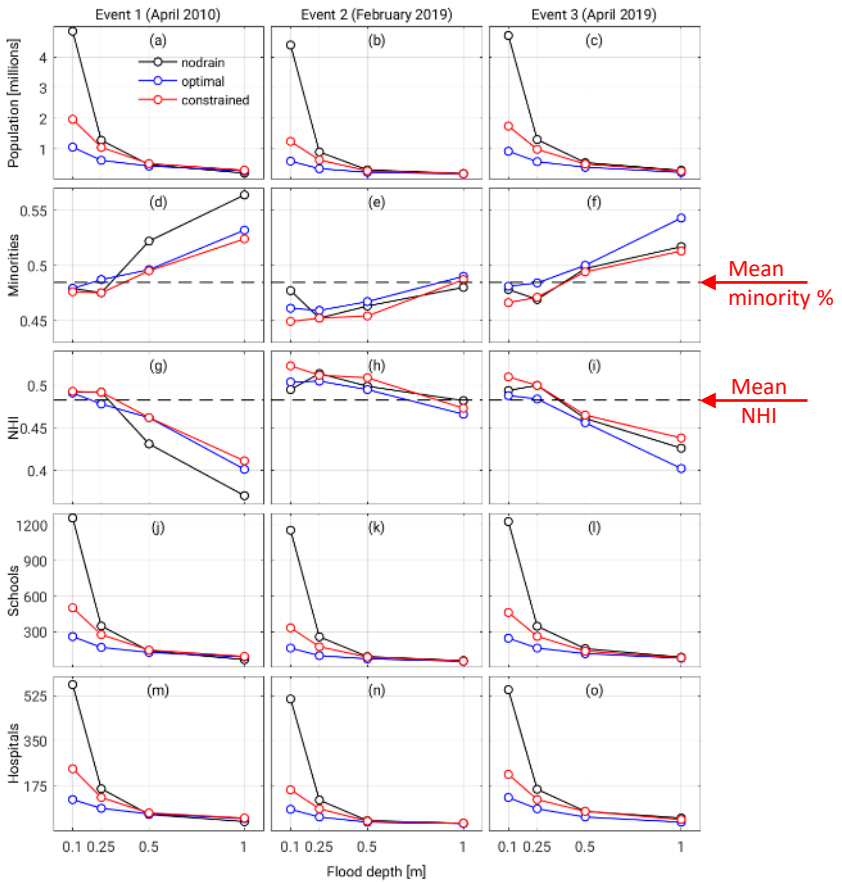
Mapping social infrastructure exposed to flooding

Additional census data shows that minorities and low-income populations are disproportionally impacted by severe floods







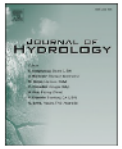


Model description and results published in December 2023

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol



Research papers

An urban drainage scheme for large-scale flood models

Augusto Getirana [a,b,*], Felipe Mandarino [c], Patricia Ney de Montezuma [d], Dalia Kirschbaum [a]

[a] Hydrological Sciences Laboratory, NASA Goddard Space Flight Center, Greenbelt, MD, United States
[b] Science Applications International Corporation, Greenbelt, MD, United States
[c] Instituto Pereira Passos, Rio de Janeiro, Brazil
[d] Rio Águas, Rio de Janeiro, Brazil

ARTICLE INFO

This manuscript was handled by Andras Bardossy, Editor-in-Chief, with the assistance of Piyatida Ruangrassamee, Associate Editor.

Keywords:
Urban drainage
Urban flood
Flood model
Flood exposure
Extreme events
Rio de Janeiro

ABSTRACT

As flood modeling spatial resolutions get finer, physical processes normally neglected, such as urban drainage, must be accounted for. Here, we describe and evaluate an urban drainage scheme for large-scale flood models. The parameterization accounts for urban imperviousness, and water flow over streets and through a prescribed urban drainage network. A parameter sensitivity analysis is performed during three major extreme floods over Rio de Janeiro city, Brazil, at ~200 m spatial resolution. Results show that, compared to a hypothetical case without urban drainage, representing a drainage network decreases urban flooding during selected extreme events across Rio de Janeiro by 31–53 %. Such a decrease is caused by an underground water storage of up to 2.5 billion m^3 across the city during flood peaks. Underground water storage and transport smooth out and delay peak flows by a few hours over major rivers and channels draining the city. Simulations also indicate that the number of residents exposed to flooding drops by 60–80 %, from ~5 million to 1–2 million, when an urban drainage system is considered during extreme events. Similar proportions are found for social infrastructure (i.e., schools and hospitals) exposed to flooding. Results reveal that racial minority and low-income populations could disproportionally be exposed to extreme floodings across the city. We conclude that representing urban drainage has a substantial impact on flood exposure and should be accounted for in fine resolution modeling. The proposed scheme is particularly useful in poorly monitored cities and where extreme floods are a frequent hazard yet to be tackled.

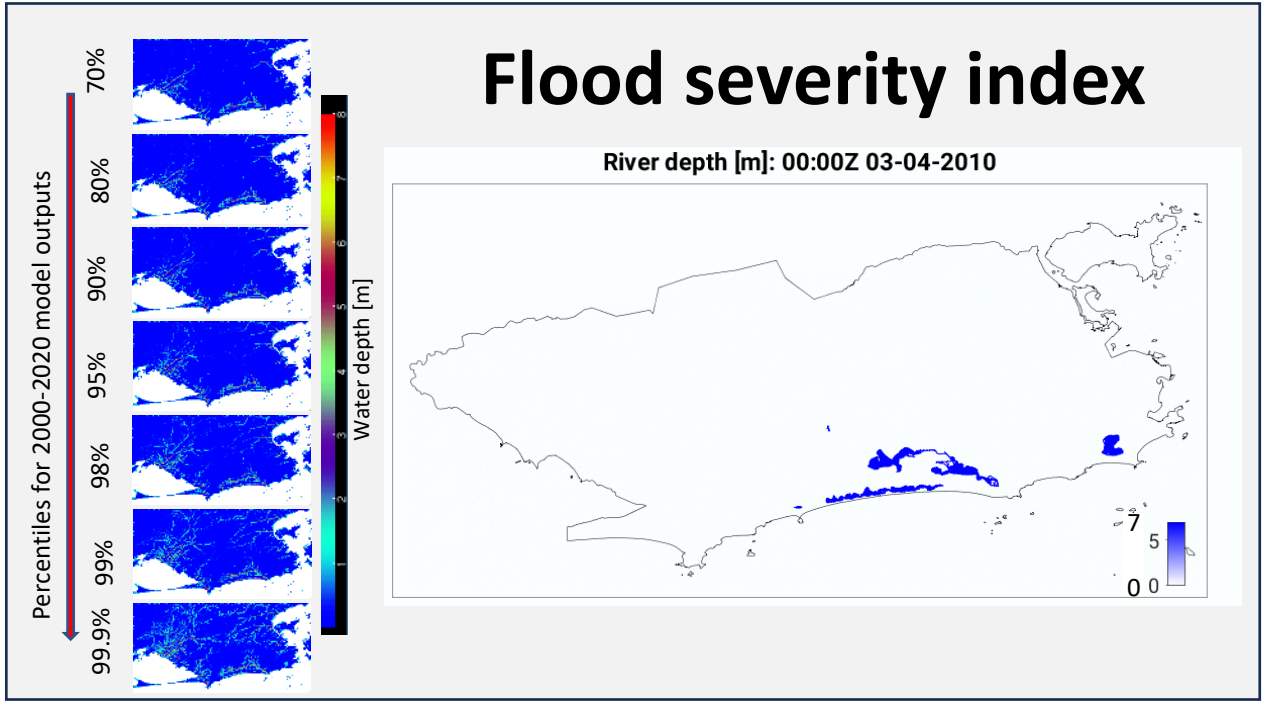
Next steps

1. Final transition to operations

- The automated system will run every 15 minutes and provide near real time flood severity index

2. Flood forecast

- Search for suitable meteorological forecasts



Content of this seminar

- Rivers and floodplains as key components of terrestrial water storage variability
- Climate and human impacts on flood risk in Southern Louisiana
- Development of a flood model for Rio de Janeiro city, Brazil
- Final considerations and prospects

Final considerations & prospects

- HyMAP has been significantly improved in the past 12 years, since its initial implementation in LIS;
- The combination of all capabilities currently available in HyMAP puts it as one of the most advanced global flood models available;
- Ongoing and future developments:
  - Surface water-groundwater interface;
  - SWOT data assimilation;
  - Improved reservoir operation scheme;
  - Inclusion of natural lakes and a water quality module.
- HyMAP is freely available in LIS through GitHub (https://github.com/NASA-LIS/LISF);
- New collaborations for either new developments and applications are always welcome!

GODDARD EARTH SCIENCES